Configuration — every knob that matters

The gates that stop live trading

Live requires both, and `Settings.is_live_armed()` enforces the AND:

```
TRADING_ENV=live
ALLOW_LIVE_TRADING=true
```

Either one alone does nothing. `trading live` refuses to start otherwise. This is the single most important invariant in the repo and nothing should ever weaken it — not a flag, not a "temporary" override, not a test fixture that patches it globally.

Where each setting actually lives

There are three config surfaces and they do not overlap the way the file names suggest.

Surface	Read by	Contains
`.env`	`core/config.Settings` (pydantic-settings)	All risk limits the risk manager enforces, secrets, broker, universe/strategy selection
`config/risk.yaml`	backtester + simulator only	Slippage model, documented intent. The portfolio/position/drawdown numbers here are NOT loaded by the risk manager.
`config/universes.yaml` + `universes.generated.yaml`	`core/universes.load_universe`	Symbol sets. Hand-curated file wins on key collision.

That second row is a trap and the file's own header now warns about it: editing `max_position_pct` in `risk.yaml` changes nothing about live risk. `.env` is the source of truth.

Risk limits (`.env`)

```
MAX_GROSS_EXPOSURE=1.0        # long-only, no leverage
MAX_POSITION_PCT=0.10         # any single name
MAX_DAILY_LOSS_PCT=0.02       # one-day kill switch
MAX_DRAWDOWN_PCT=0.15         # full halt until a human resets
MAX_MARGIN_BORROWING_PCT=...
REQUIRE_CYCLE_APPROVAL=       # human confirms each rebalance
CYCLE_APPROVAL_TIMEOUT_S=
```

What trades and where

```
UNIVERSE=sp500                # also drives the agents' candidate ladder
STRATEGY=top_k_momentum
CRON=                         # rebalance schedule
```

`UNIVERSE` and `STRATEGY` are shared between the trading loop and the agent candidate ladder on purpose — the agents rank what the system trades, so the ladder is the house view rather than a second opinion.

Model routing

```
AGENTS_ENABLED=true
ANTHROPIC_API_KEY=...
AGENTS_MODEL=                # specialists; default claude-sonnet-4-6
AGENTS_MODEL_FRONTIER=       # decision nodes; default claude-opus-4-8
```

Three nodes run on the frontier model: challenger, manager, and the agent PM. Everything else is mid-tier. Opus is therefore in the per-cycle path — watch credit burn. A June 2026 committee outage was simply zero API credit (Anthropic returns HTTP 400).

Hard caps that live in code, not config

These are in `agents/pm.py` and bind the simulated sleeve. They are deliberately code rather than YAML: they are invariants, not tunables.

Constant	Value	Why
`MAX_WEIGHT_PER_NAME`	0.25	ETFs are diversified
`MAX_WEIGHT_PER_STOCK`	0.10	single names carry idiosyncratic risk
`MAX_CLUSTER`	0.50	six 0.9-correlated names is one bet wearing six hats
`MAX_ETF_POSITIONS`	3	forces individual-name expression
`MAX_GROSS`	1.0	long-only
`STALE_BOOK_CYCLES`	3	cycles of an unchanged name set before the digest complains
`COST_BPS`	10	commission + slippage charged on sim turnover

`CLUSTERS` in the same file maps ~60 tickers to five correlated groups (tech_complex, energy, health, financials, defense). It is deliberately beta-based rather than GICS-pure — AMZN and TSLA trade with tech beta, so for risk purposes they live in the tech complex. Symbols not in the map are uncapped at cluster level, which is a real gap as the universe widens.

Operator controls (Telegram)

Command	Effect	Durability
`/hold SYM`	Freezes a position out of the rebalance; PM is hard-blocked from allocating to it	until `/unhold`
natural-language instruction	Captured as a mandate, graded strong/medium/soft by phrasing	expires
`/lesson <text>`	Written to permanent memory; firm tone → `established`, else `candidate`	permanent
`/lessons harden|soften <id>`	Re-weight a lesson	permanent
`/halt`, `/resume`	Kill switch	until reset
`/approve`, `/reject`, `/pick`	Gate a pending cycle	per cycle

Mandates and lessons are different things and the difference matters: a mandate is an instruction for the next run that is then consumed; a lesson is a durable belief that agents carry forward and that can be graded. See `04_PRODUCTION_READINESS.md` for the current gap in how mandates are detected.